TidyTuesday Week 45

Pranay Gundam

Sunday 8th November, 2026

Contents

1 Weekly Summary **2**

2 Date: 2026-11-02 **4**
- 2.1 Raw Regression . . . 4
- 2.2 Detrended Regression . . . 6

3 Date: 2026-11-03 **8**
- 3.1 Raw Regression . . . 8
- 3.2 Detrended Regression . . . 11

4 Date: 2026-11-04 **13**
- 4.1 Raw Regression . . . 13
- 4.2 Detrended Regression . . . 15

5 Date: 2026-11-05 **17**
- 5.1 Raw Regression . . . 17
- 5.2 Detrended Regression . . . 19

6 Date: 2026-11-06 **21**
- 6.1 Raw Regression . . . 21
- 6.2 Detrended Regression . . . 24

7 Date: 2026-11-07 **26**
- 7.1 Raw Regression . . . 26
- 7.2 Detrended Regression . . . 28

8 Date: 2026-11-08 **30**
- 8.1 Raw Regression . . . 30
- 8.2 Detrended Regression . . . 32

1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to another thrilling installment of Pulling Two Random Numbers Out of a Government Database and Shoving Them Into a Line. As always, the premise is as noble as it is pointless: every day we yank a fresh pair of unrelated economic series from FRED, draw a straight line between them, and then act surprised when the line fits. The whole project is a monument to spurious correlation, so please do not use anything below to, say, run a hedge fund or win an argument with your uncle.

We kicked off Sunday with goods-producing employment in Springfield, Massachusetts versus a nursing home price index, which posted a gaudy raw R-squared of 0.78 – the kind of number that makes you want to pop champagne. Then we detrended it and the relationship evaporated into literally nothing (R-squared 0.02, p-value basically 0.52, which is statistics-speak for "lol no"). Textbook trend mirage: both series drifted over the years, the regression noticed the drift, and we almost fell for it. Monday's IT manufacturing orders versus Missouri bank equity returns was more honest about being weak – a modest raw fit that limped down but at least stayed technically significant after detrending, which is the equivalent of a participation trophy. Tuesday pitted East North Central residential electricity prices against restaurant sales, and here's where things got mildly interesting: the raw fit was strong (0.74), and even after stripping the trends out it held on to a respectable 0.21 with a p-value that refuses to be ignored. I'm not saying your dinner bill is powered by the electric company, but I'm also not not saying it.

The real headline-grabbers came midweek. Wednesday's matchup – Puerto Rico covered employment versus U.S. travel service exports – was the inverse of the usual gag: the raw relationship was a shrug (R-squared 0.06, not even significant), but detrending the thing somehow cranked it UP to 0.52 with a p-value sporting eight zeros. A relationship that only appears once you remove the trends is a genuinely rare and spooky little creature, and I have no earthly idea what it means. Friday gave us the week's crown jewel: Norwegian recession indicators versus Northeast adjustable-rate mortgage margins, which held strong in both the raw AND detrended versions (R-squared climbing from 0.17 to 0.22, p-values so small they need scientific notation to even exist). Two discontinued series from different countries, both dead for years, reaching out from the grave to hold hands. Beautiful. Meaningless, almost certainly, but beautiful.

Rounding out the week, Thursday matched the trade balance against Midwest chicken breast prices, posting a solid raw fit that survived detrending in weakened form – a mild "huh" rather than a full gasp. And Saturday closed us out with West Coast banana prices versus Mideast machinery manufacturing output, which looked promising raw (0.45) and then face-planted com-

pletely once detrended (0.02, p-value 0.52), joining the Springfield special in this week's Hall of Trend Mirages. So the final tally: a couple of fun survivors, a couple of spectacular collapses, and absolutely zero reason to believe any of it. See you next week for more rigorous nonsense.

2 Date: 2026-11-02

Series ID: SMU25781000600000001
This series is titled All Employees: Goods Producing in Springfield, MA-CT (NECTA) and has a frequency of Monthly. The units are Thousands of Persons and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1990-01-01 and the observation end date is 2024-12-01.The popularity of this series is 1.

Series ID: PRGONUPIHCSA
This series is titled Medical Services Expenditures by Provider: Nursing Homes: Proprietary and Government Nursing Homes Price Index and has a frequency of Annual. The units are Index 2017=100 and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 2000-01-01 and the observation end date is 2021-01-01.The popularity of this series is 1.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_PRGONUPIHCSA	**R-squared:**	0.783
Model:	OLS	**Adj. R-squared:**	0.772
Method:	Least Squares	**F-statistic:**	72.01
Date:	Mon, 02 Nov 2026	**Prob (F-statistic):**	4.64e-08
Time:	21:06:51	**Log-Likelihood:**	-73.979
No. Observations:	22	**AIC:**	152.0
Df Residuals:	20	**BIC:**	154.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	186.2043	11.787	15.797	0.000	161.616	210.793
value_fred_SMU25781000600000001	-2.2652	0.267	-8.486	0.000	-2.822	-1.708

Omnibus:	5.822	**Durbin-Watson:**	0.251
Prob(Omnibus):	0.054	**Jarque-Bera (JB):**	4.103
Skew:	1.043	**Prob(JB):**	0.129
Kurtosis:	3.353	**Cond. No.**	333.

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

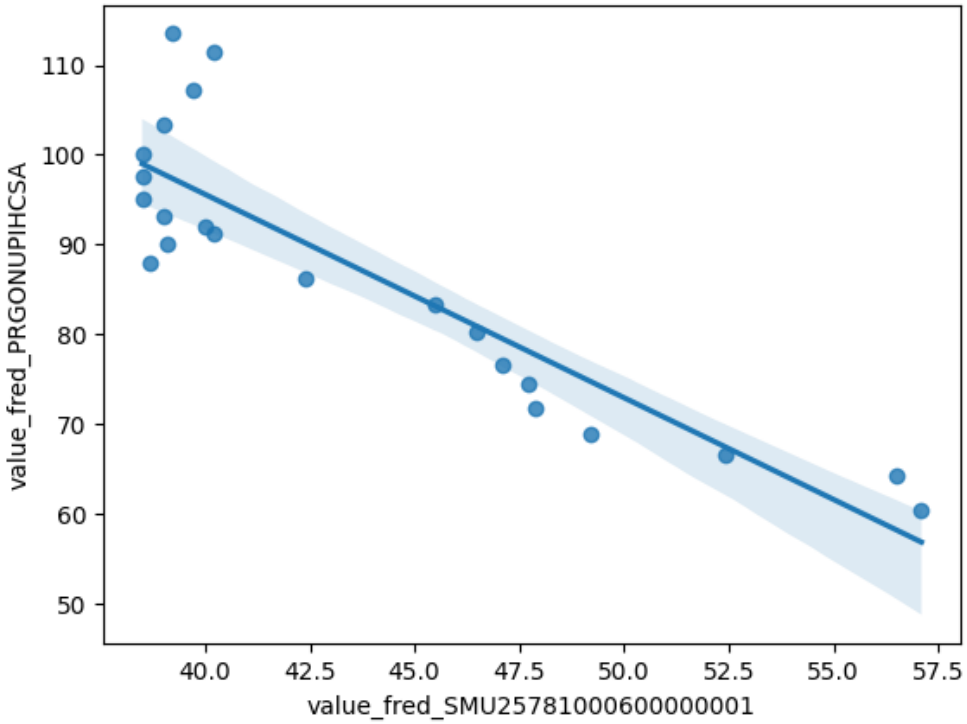


Figure 1: Raw Regression Plot for 2026-11-02

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_PRGONUPIHCSA_detrended	**R-squared:**	0.021
Model:	OLS	**Adj. R-squared:**	-0.028
Method:	Least Squares	**F-statistic:**	0.4223
Date:	Mon, 02 Nov 2026	**Prob (F-statistic):**	0.523
Time:	21:06:51	**Log-Likelihood:**	-42.096
No. Observations:	22	**AIC:**	88.19
Df Residuals:	20	**BIC:**	90.37
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	4.027e-13	0.367	1.1e-12	1.000	-0.765	0.765
value_fred_SMU25781000600000001_detrended	-0.0866	0.133	-0.650	0.523	-0.365	0.191

Omnibus:	2.262	**Durbin-Watson:**	0.331
Prob(Omnibus):	0.323	**Jarque-Bera (JB):**	1.135
Skew:	-0.097	**Prob(JB):**	0.567
Kurtosis:	1.904	**Cond. No.**	2.75

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

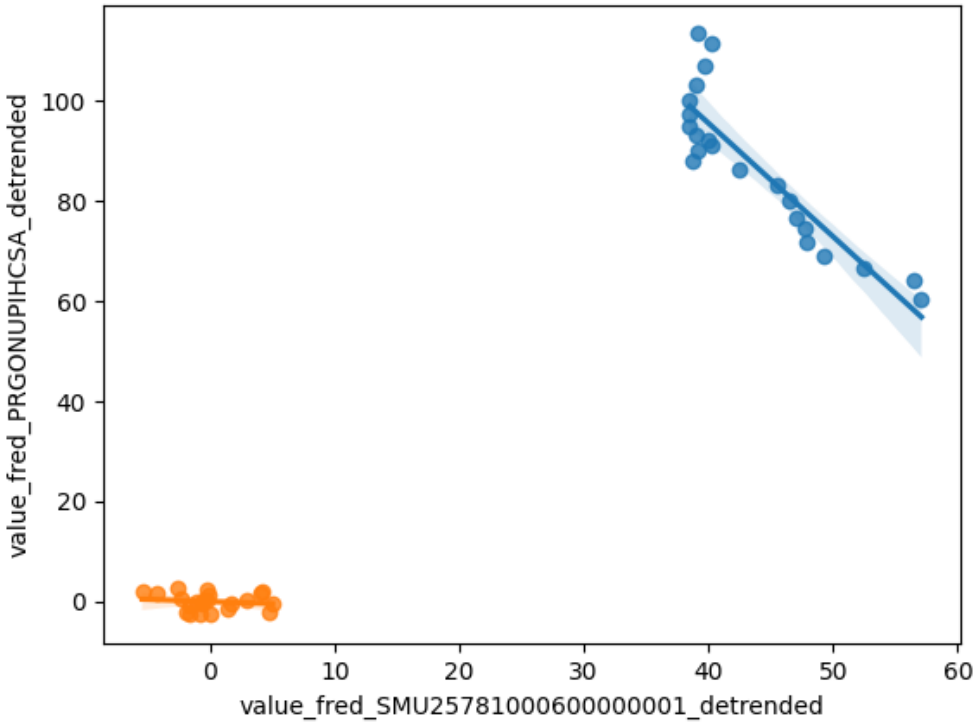


Figure 2: Detrended Regression Plot for 2026-11-02

3 Date: 2026-11-03

Series ID: AITINO
This series is titled Manufacturers' New Orders: Information Technology Industries and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Seasonally Adjusted.The observation start date is 1992-02-01 and the observation end date is 2026-09-01.The popularity of this series is 14.

Series ID: MOROE
This series is titled Return on Average Equity for Banks in Missouri (DISCONTINUED) and has a frequency of Quarterly, End of Period. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1984-01-01 and the observation end date is 2020-07-01.The popularity of this series is 1.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_MOROE	**R-squared:**	0.135
Model:	OLS	**Adj. R-squared:**	0.128
Method:	Least Squares	**F-statistic:**	17.51
Date:	Tue, 03 Nov 2026	**Prob (F-statistic):**	5.69e-05
Time:	18:56:19	**Log-Likelihood:**	-307.15
No. Observations:	114	**AIC:**	618.3
Df Residuals:	112	**BIC:**	623.8
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	1.9234	2.332	0.825	0.411	-2.697	6.544
value_fred_AITINO	0.0004	9.45e-05	4.185	0.000	0.000	0.001

Omnibus:	41.602	**Durbin-Watson:**	0.161
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	88.948
Skew:	-1.485	**Prob(JB):**	4.84e-20
Kurtosis:	6.147	**Cond. No.**	1.70e+05

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

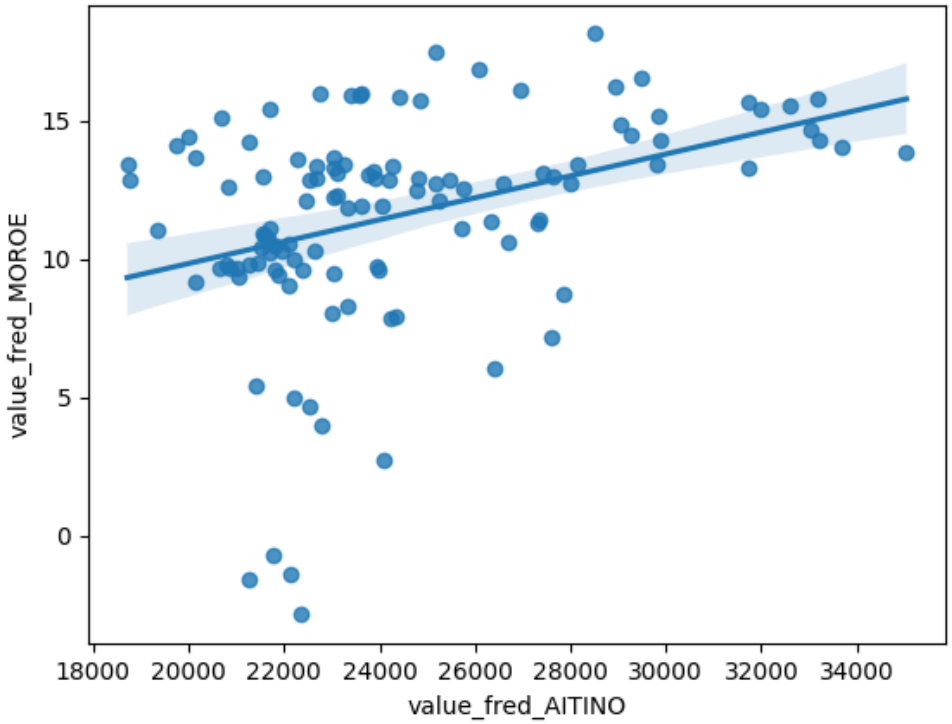


Figure 3: Raw Regression Plot for 2026-11-03

[2] The condition number is large, 1.7e+05. This might indicate that there are strong multicollinearity or other numerical problems.

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_MOROE_detrended	**R-squared:**	0.057
Model:	OLS	**Adj. R-squared:**	0.049
Method:	Least Squares	**F-statistic:**	6.779
Date:	Tue, 03 Nov 2026	**Prob (F-statistic):**	0.0105
Time:	18:56:19	**Log-Likelihood:**	-295.86
No. Observations:	114	**AIC:**	595.7
Df Residuals:	112	**BIC:**	601.2
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-7.674e-13	0.306	-2.5e-12	1.000	-0.607	0.607
value_fred_AITINO_detrended	0.0002	9.13e-05	2.604	0.010	5.68e-05	0.000

Omnibus:	60.278	**Durbin-Watson:**	0.190
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	192.530
Skew:	-1.974	**Prob(JB):**	1.56e-42
Kurtosis:	7.994	**Cond. No.**	3.35e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 3.35e+03. This might indicate that there are strong multicollinearity or other numerical problems.

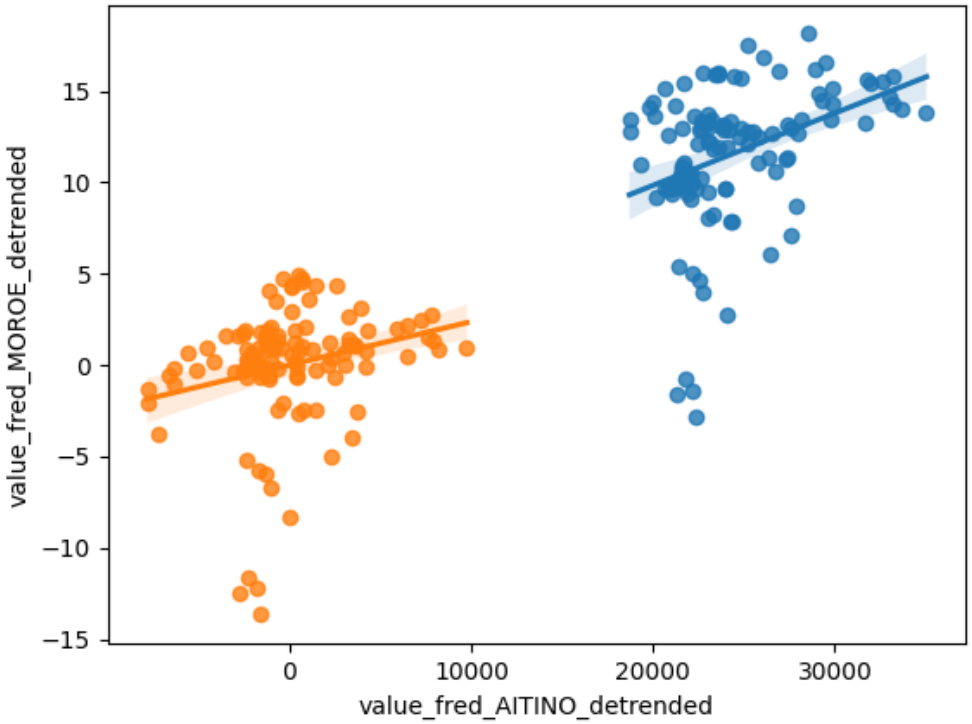


Figure 4: Detrended Regression Plot for 2026-11-03

4 Date: 2026-11-04

Series ID: WPU054121013
This series is titled Producer Price Index by Commodity: Fuels and Related Products and Power: Residential Electric Power for East North Central Census Division and has a frequency of Monthly. The units are Index Dec 2007=100 and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 2007-12-01 and the observation end date is 2026-09-01.The popularity of this series is 1.

Series ID: MRTSSM722USN
This series is titled Retail Sales: Food Services and Drinking Places and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1992-01-01 and the observation end date is 2026-08-01.The popularity of this series is 19.

4.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_MRTSSM722USN	**R-squared:**	0.737
Model:	OLS	**Adj. R-squared:**	0.734
Method:	Least Squares	**F-statistic:**	318.7
Date:	Wed, 04 Nov 2026	**Prob (F-statistic):**	8.31e-35
Time:	22:07:01	**Log-Likelihood:**	-1192.6
No. Observations:	116	**AIC:**	2389.
Df Residuals:	114	**BIC:**	2395.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-2.164e+04	5164.973	-4.190	0.000	-3.19e+04	-1.14e+04
value_fred_WPU054121013	575.5062	32.238	17.852	0.000	511.643	639.370

Omnibus:	8.963	**Durbin-Watson:**	0.814
Prob(Omnibus):	0.011	**Jarque-Bera (JB):**	10.211
Skew:	-0.481	**Prob(JB):**	0.00606
Kurtosis:	4.089	**Cond. No.**	1.25e+03

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

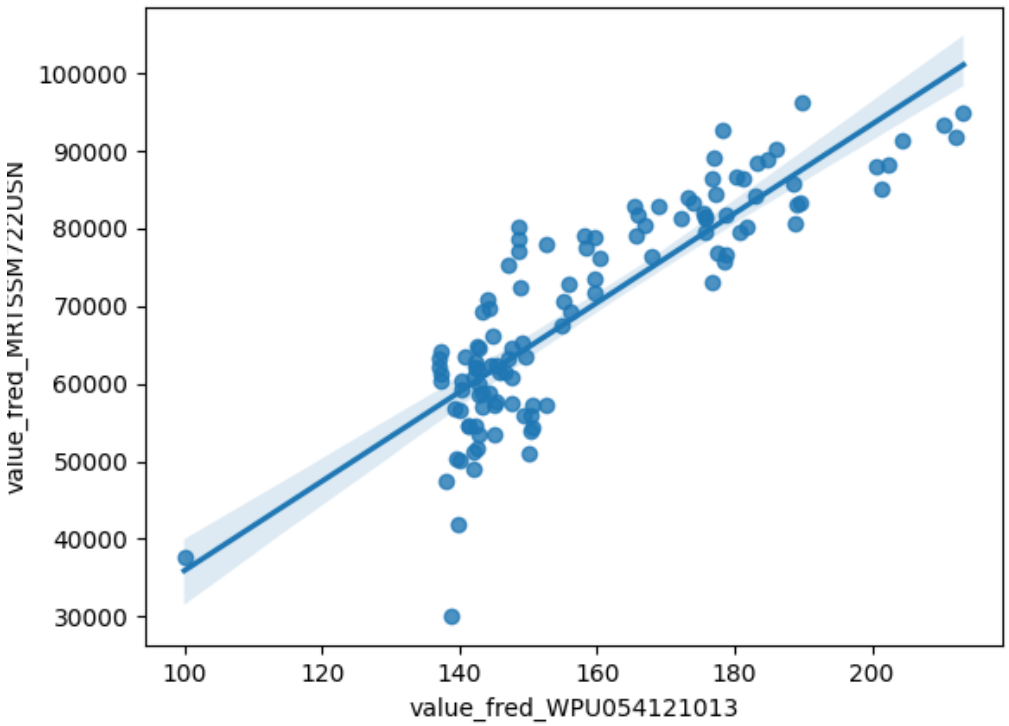


Figure 5: Raw Regression Plot for 2026-11-04

[2] The condition number is large, 1.25e+03. This might indicate that there are strong multicollinearity or other numerical problems.

4.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_MRTSSM722USN_detrended	**R-squared:**	0.211
Model:	OLS	**Adj. R-squared:**	0.204
Method:	Least Squares	**F-statistic:**	30.56
Date:	Wed, 04 Nov 2026	**Prob (F-statistic):**	2.08e-07
Time:	22:07:01	**Log-Likelihood:**	-1180.4
No. Observations:	116	**AIC:**	2365.
Df Residuals:	114	**BIC:**	2370.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	1.527e-09	595.106	2.57e-12	1.000	-1178.899	1178.899
value_fred_WPU054121013_detrended	318.6043	57.635	5.528	0.000	204.429	432.779

Omnibus:	41.308	**Durbin-Watson:**	0.835
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	110.393
Skew:	-1.327	**Prob(JB):**	1.07e-24
Kurtosis:	6.975	**Cond. No.**	10.3

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

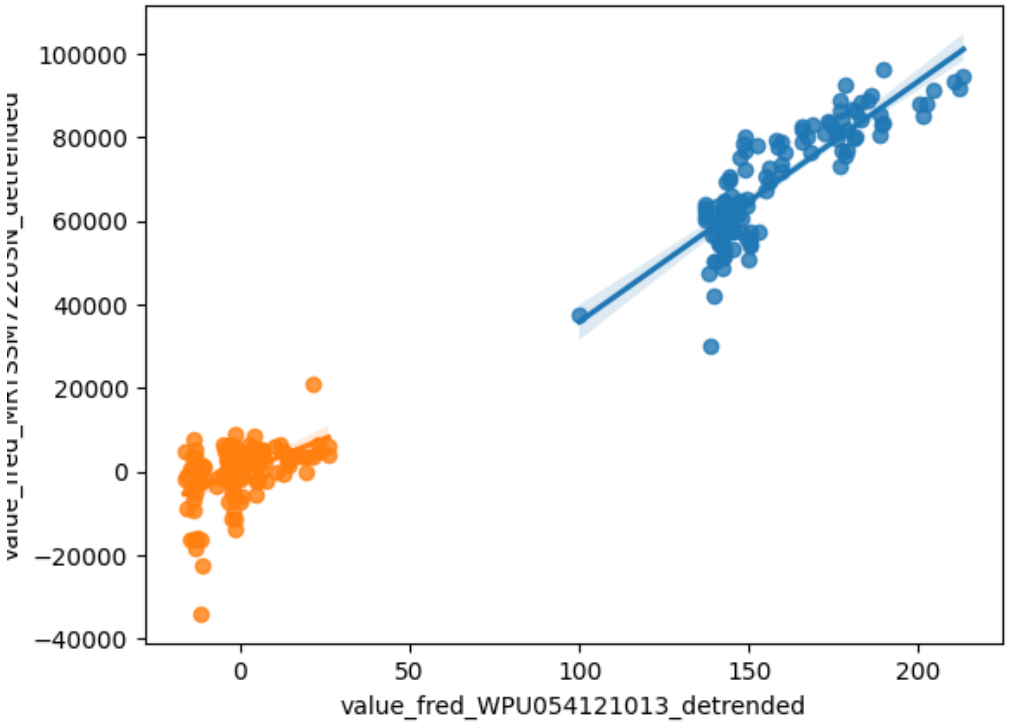


Figure 6: Detrended Regression Plot for 2026-11-04

5 Date: 2026-11-05

Series ID: PRICEMPLOY
This series is titled Covered Employment in Puerto Rico and has a frequency of Weekly, Ending Saturday. The units are Number and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1986-02-08 and the observation end date is 2026-10-17.The popularity of this series is 1.

Series ID: BOXTVLM133S
This series is titled U.S. Exports of Services - Travel and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Seasonally Adjusted.The observation start date is 1992-01-01 and the observation end date is 2017-09-01.The popularity of this series is 2.

5.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_BOXTVLM133S	**R-squared:**	0.064
Model:	OLS	**Adj. R-squared:**	0.042
Method:	Least Squares	**F-statistic:**	2.938
Date:	Thu, 05 Nov 2026	**Prob (F-statistic):**	0.0937
Time:	19:20:26	**Log-Likelihood:**	-420.46
No. Observations:	45	**AIC:**	844.9
Df Residuals:	43	**BIC:**	848.5
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	1.907e+04	6693.471	2.850	0.007	5574.413	3.26e+04
value_fred_PRICEMPLOY	-0.0121	0.007	-1.714	0.094	-0.026	0.002

Omnibus:	5.659	**Durbin-Watson:**	0.056
Prob(Omnibus):	0.059	**Jarque-Bera (JB):**	2.550
Skew:	0.283	**Prob(JB):**	0.279
Kurtosis:	1.981	**Cond. No.**	1.51e+07

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.51e+07. This might indicate that there are
strong multicollinearity or other numerical problems.

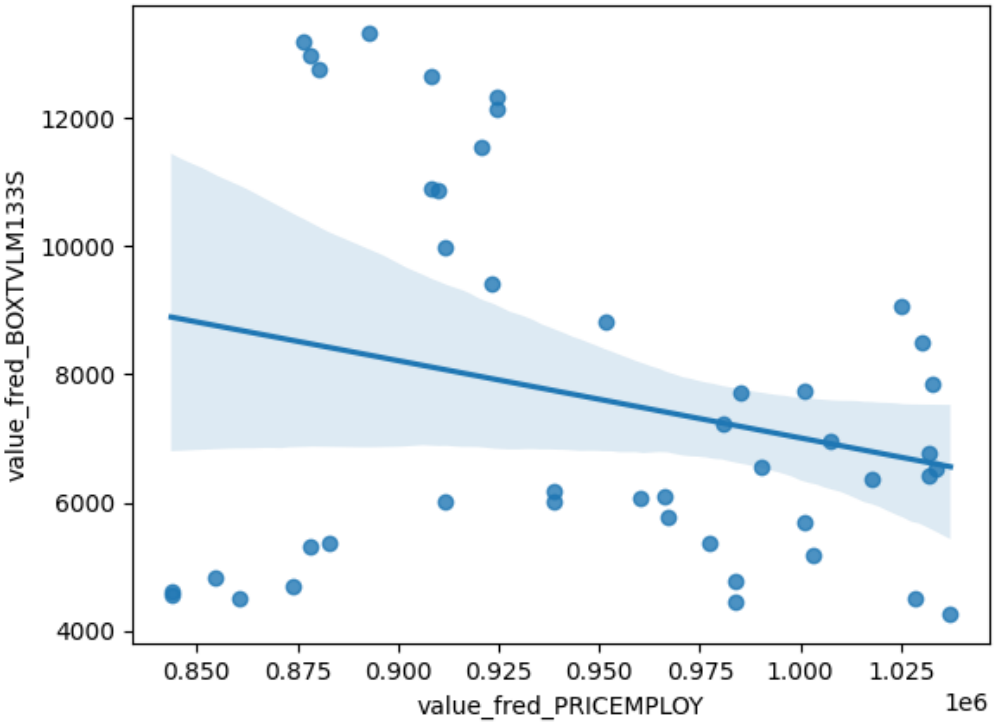


Figure 7: Raw Regression Plot for 2026-11-05

5.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_BOXTVLM133S_detrended	**R-squared:**	0.519
Model:	OLS	**Adj. R-squared:**	0.508
Method:	Least Squares	**F-statistic:**	46.42
Date:	Thu, 05 Nov 2026	**Prob (F-statistic):**	2.40e-08
Time:	19:20:26	**Log-Likelihood:**	-369.72
No. Observations:	45	**AIC:**	743.4
Df Residuals:	43	**BIC:**	747.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-2.659e-10	136.526	-1.95e-12	1.000	-275.330	275.330
value_fred_PRICEMPLOY_detrended	-0.0156	0.002	-6.813	0.000	-0.020	-0.011

Omnibus:	0.541	**Durbin-Watson:**	0.520
Prob(Omnibus):	0.763	**Jarque-Bera (JB):**	0.636
Skew:	-0.013	**Prob(JB):**	0.727
Kurtosis:	2.418	**Cond. No.**	5.97e+04

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 5.97e+04. This might indicate that there are strong multicollinearity or other numerical problems.

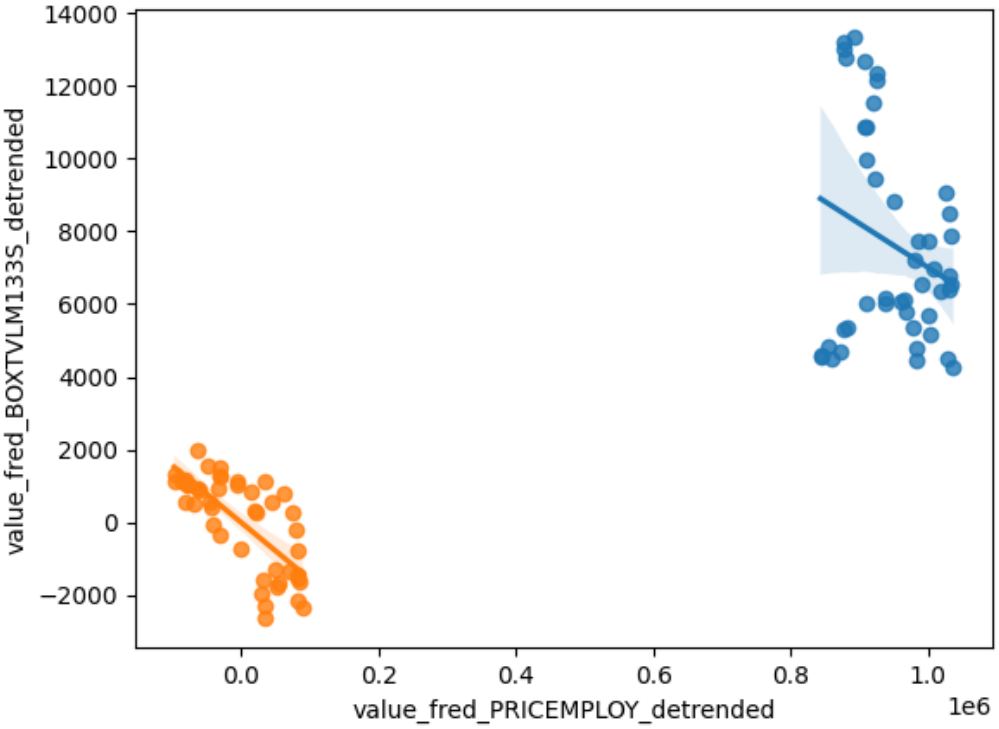


Figure 8: Detrended Regression Plot for 2026-11-05

6 Date: 2026-11-06

Series ID: IEABCGSN
This series is titled Balance on goods and services and has a frequency of Quarterly. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1999-01-01 and the observation end date is 2026-04-01.The popularity of this series is 3.

Series ID: APU0200FF1101
This series is titled Average Price: Chicken Breast, Boneless (Cost per Pound/453.6 Grams) in the Midwest Census Region - Urban and has a frequency of Monthly. The units are U.S. Dollars and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 2006-01-01 and the observation end date is 2026-09-01.The popularity of this series is 3.

6.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_APU0200FF1101	**R-squared:**	0.277
Model:	OLS	**Adj. R-squared:**	0.268
Method:	Least Squares	**F-statistic:**	30.27
Date:	Fri, 06 Nov 2026	**Prob (F-statistic):**	4.50e-07
Time:	18:35:36	**Log-Likelihood:**	-61.425
No. Observations:	81	**AIC:**	126.8
Df Residuals:	79	**BIC:**	131.6
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	2.5029	0.195	12.831	0.000	2.115	2.891
value_fred_IEABCGSN	-6.612e-06	1.2e-06	-5.502	0.000	-9e-06	-4.22e-06

Omnibus:	17.392	**Durbin-Watson:**	0.410
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	20.526
Skew:	1.099	**Prob(JB):**	3.49e-05
Kurtosis:	4.119	**Cond. No.**	5.45e+05

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

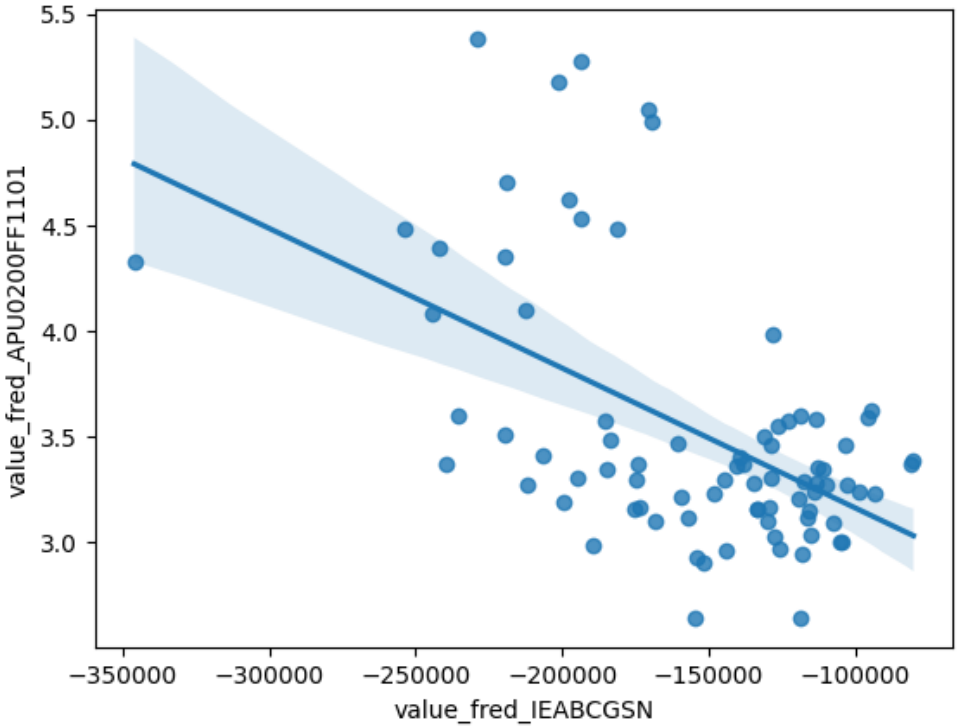


Figure 9: Raw Regression Plot for 2026-11-06

[2] The condition number is large, 5.45e+05. This might indicate that there are strong multicollinearity or other numerical problems.

6.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_APU0200FF1101_detrended	**R-squared:**	0.144
Model:	OLS	**Adj. R-squared:**	0.134
Method:	Least Squares	**F-statistic:**	13.33
Date:	Fri, 06 Nov 2026	**Prob (F-statistic):**	0.000467
Time:	18:35:36	**Log-Likelihood:**	-54.751
No. Observations:	81	**AIC:**	113.5
Df Residuals:	79	**BIC:**	118.3
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	3.741e-14	0.054	6.99e-13	1.000	-0.107	0.107
value_fred_IEABCGSN_detrended	-4.522e-06	1.24e-06	-3.651	0.000	-6.99e-06	-2.06e-06

Omnibus:	6.297	**Durbin-Watson:**	0.365
Prob(Omnibus):	0.043	**Jarque-Bera (JB):**	5.617
Skew:	0.545	**Prob(JB):**	0.0603
Kurtosis:	3.690	**Cond. No.**	4.32e+04

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 4.32e+04. This might indicate that there are strong multicollinearity or other numerical problems.

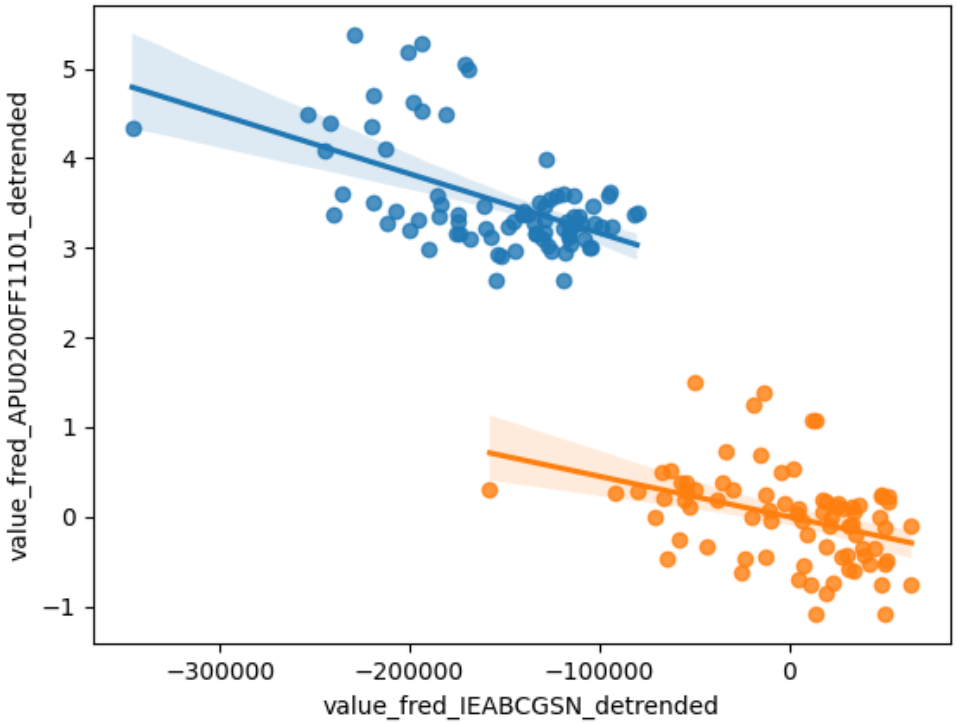


Figure 10: Detrended Regression Plot for 2026-11-06

7 Date: 2026-11-07

Series ID: NORRECDM
This series is titled OECD based Recession Indicators for Norway from the Peak through the Trough (DISCONTINUED) and has a frequency of Daily, 7-Day. The units are +1 or 0 and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1960-02-01 and the observation end date is 2022-09-30.The popularity of this series is 2.

Series ID: MORTMRGN5NE
This series is titled Margin for 5/1-Year Adjustable Rate Mortgage in the Northeast Freddie Mac Region (DISCONTINUED) and has a frequency of Weekly, Ending Thursday. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 2005-01-06 and the observation end date is 2015-12-31.The popularity of this series is 1.

7.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_MORTMRGN5NE	**R-squared:**	0.174
Model:	OLS	**Adj. R-squared:**	0.172
Method:	Least Squares	**F-statistic:**	120.2
Date:	Sat, 07 Nov 2026	**Prob (F-statistic):**	1.63e-25
Time:	17:03:06	**Log-Likelihood:**	1072.5
No. Observations:	574	**AIC:**	-2141.
Df Residuals:	572	**BIC:**	-2132.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	2.7734	0.002	1388.819	0.000	2.769	2.777
value_fred_NORRECDM	-0.0351	0.003	-10.963	0.000	-0.041	-0.029

Omnibus:	54.533	**Durbin-Watson:**	0.276
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	314.645
Skew:	0.052	**Prob(JB):**	4.74e-69
Kurtosis:	6.626	**Cond. No.**	2.44

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

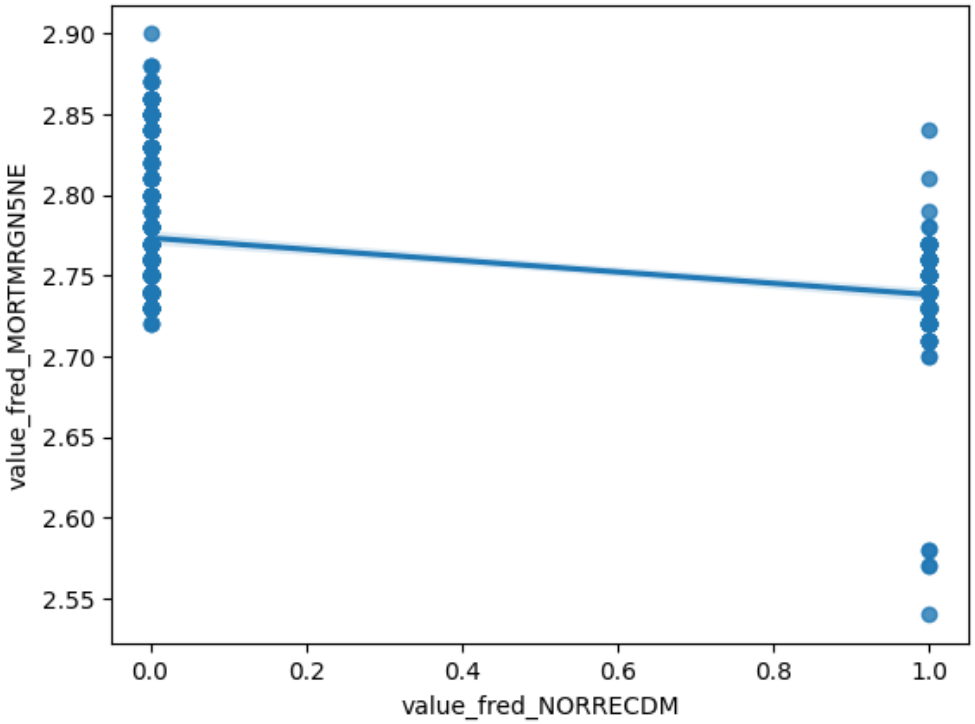


Figure 11: Raw Regression Plot for 2026-11-07

7.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_MORTMRGN5NE_detrended	**R-squared:**	0.220
Model:	OLS	**Adj. R-squared:**	0.219
Method:	Least Squares	**F-statistic:**	161.6
Date:	Sat, 07 Nov 2026	**Prob (F-statistic):**	8.78e-33
Time:	17:03:06	**Log-Likelihood:**	1186.5
No. Observations:	574	**AIC:**	-2369.
Df Residuals:	572	**BIC:**	-2360.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	2.243e-13	0.001	1.75e-10	1.000	-0.003	0.003
value_fred_NORRECDM_detrended	-0.0334	0.003	-12.712	0.000	-0.039	-0.028

Omnibus:	209.063	**Durbin-Watson:**	0.410
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	2385.406
Skew:	-1.266	**Prob(JB):**	0.00
Kurtosis:	12.661	**Cond. No.**	2.05

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

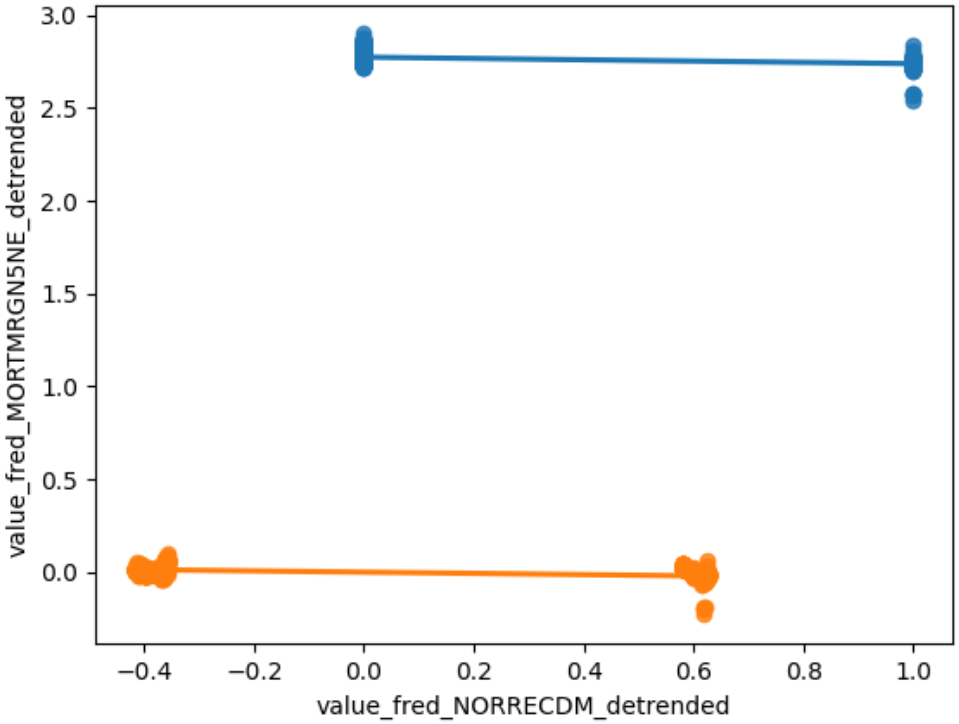


Figure 12: Detrended Regression Plot for 2026-11-07

8 Date: 2026-11-08

Series ID: APU0400711211
This series is titled Average Price: Bananas (Cost per Pound/453.6 Grams) in the West Census Region - Urban and has a frequency of Monthly. The units are U.S. Dollars and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1980-01-01 and the observation end date is 2026-09-01.The popularity of this series is 2.

Series ID: MESTMACHMANQGSP
This series is titled Chain-Type Quantity Index for Real GDP: Machinery Manufacturing (333) in the Mideast BEA Region and has a frequency of Annual. The units are Index 2017=100 and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1997-01-01 and the observation end date is 2025-01-01.The popularity of this series is 1.

8.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_MESTMACHMANQGSP	**R-squared:**	0.445
Model:	OLS	**Adj. R-squared:**	0.425
Method:	Least Squares	**F-statistic:**	21.68
Date:	Sun, 08 Nov 2026	**Prob (F-statistic):**	7.67e-05
Time:	17:18:00	**Log-Likelihood:**	-113.18
No. Observations:	29	**AIC:**	230.4
Df Residuals:	27	**BIC:**	233.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	200.5465	20.614	9.729	0.000	158.250	242.843
value_fred_APU0400711211	-155.5450	33.406	-4.656	0.000	-224.088	-87.002

Omnibus:	2.716	**Durbin-Watson:**	0.630
Prob(Omnibus):	0.257	**Jarque-Bera (JB):**	1.932
Skew:	0.632	**Prob(JB):**	0.381
Kurtosis:	3.011	**Cond. No.**	19.9

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

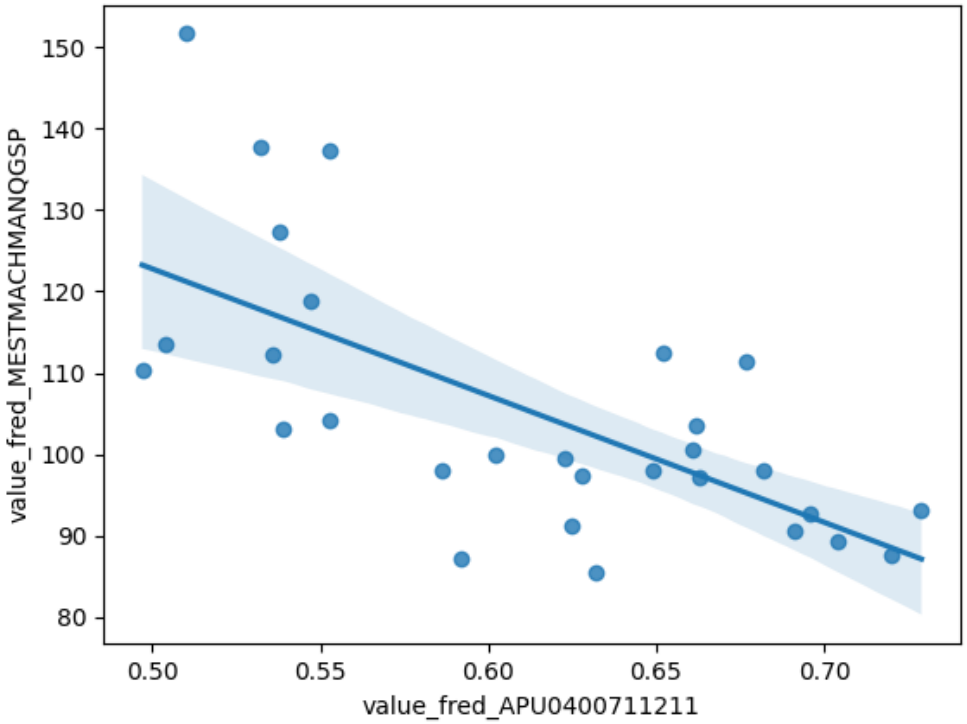


Figure 13: Raw Regression Plot for 2026-11-08

8.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_MESTMACHMANQGSP_detrended	**R-squared:**	0.015
Model:	OLS	**Adj. R-squared:**	-0.021
Method:	Least Squares	**F-statistic:**	0.4153
Date:	Sun, 08 Nov 2026	**Prob (F-statistic):**	0.525
Time:	17:18:00	**Log-Likelihood:**	-109.86
No. Observations:	29	**AIC:**	223.7
Df Residuals:	27	**BIC:**	226.5
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-2.54e-12	2.058	-1.23e-12	1.000	-4.222	4.222
value_fred_APU0400711211_detrended	-35.1716	54.578	-0.644	0.525	-147.156	76.813

Omnibus:	2.209	**Durbin-Watson:**	0.735
Prob(Omnibus):	0.331	**Jarque-Bera (JB):**	1.004
Skew:	-0.114	**Prob(JB):**	0.605
Kurtosis:	3.883	**Cond. No.**	26.5

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

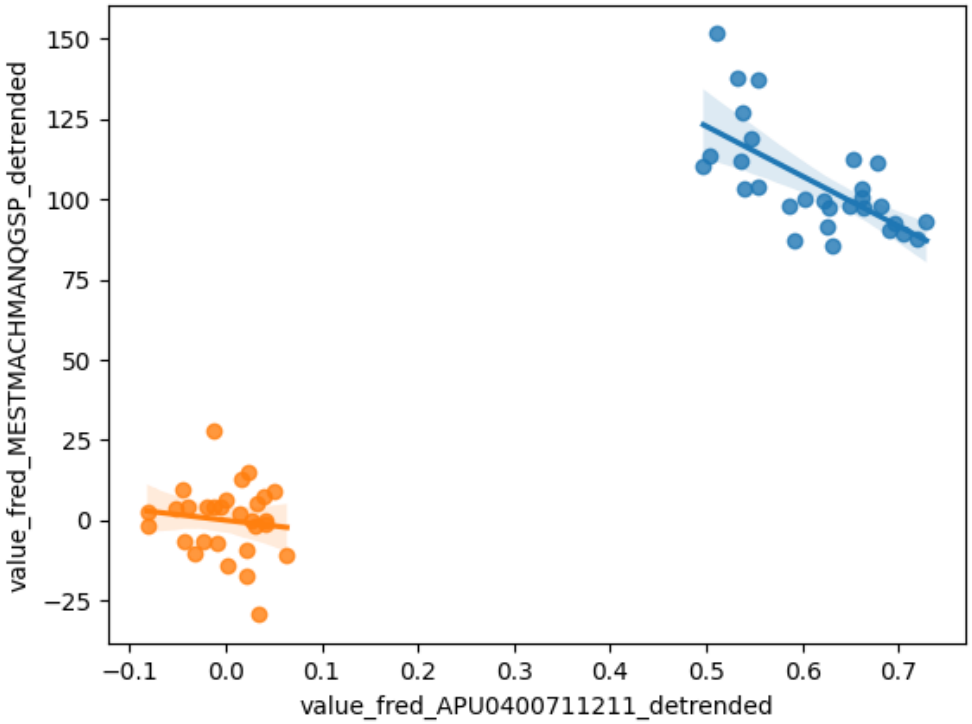


Figure 14: Detrended Regression Plot for 2026-11-08